



Machine Learning for Sustainable Energy Transitions in Africa: Evidence from Macroeconomic, Fiscal, and Environmental Drivers

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Abstract: This article analyses the energy transition trajectories of 25 African countries, combining machine learning and econometric methods. Initially, a Dynamic Time Warping (DTW)-based partitioning approach is used to divide countries into three groups with distinct socio-economic and energy profiles. The Light Gradient Boosting Machine (LightGBM) model is then used to evaluate the significance of macroeconomic and structural variables. A Panel Autoregressive Distributed Lag (Panel ARDL) model is then applied to each group to examine the short- and long-term relationships between macroeconomic and structural variables and renewable energy consumption. The results demonstrate consistency in the importance of variables identified by a Long Short-Term Memory (LSTM) model and their long-term effects within the Panel ARDL framework, thereby showcasing the robustness of the approach. The analysis reveals different dynamics: the first group is hindered by macroeconomic vulnerabilities such as financial instability and high debt, whereas the second group enjoys more favourable conditions. These results provide a basis for developing policies tailored to the specific contexts of each group to accelerate the energy transition in Africa. Thus, the study contributes to a better understanding of the key factors and helps to guide sustainable development strategies on the continent.

Keywords: Renewable energy; Clustering; Machine learning; Macroeconomic dynamics; Public debt; Urbanisation

1 Introduction

The transition toward renewable energy has become a central pillar of sustainable development strategies worldwide. Growing concerns related to climate change, energy security, and environmental degradation have accelerated the deployment of renewable energy technologies in both developed and developing economies.

Renewable energy plays a crucial role in reducing CO₂ emissions and improving environmental sustainability. Unlike fossil fuels, renewable energy sources produce little or no greenhouse gas emissions, which helps mitigate climate change and environmental degradation. Several studies have shown that increasing renewable energy consumption contributes to environmental quality improvement and climate change mitigation [1–4]. Therefore, promoting renewable energy is considered an important policy tool for achieving sustainable development, environmental protection, and long-term energy security.

Over the past decade, the rapid decline in the costs of solar and wind technologies, combined with international climate commitments, has significantly boosted investments in renewable energy [5]. However, despite this global momentum, the adoption of renewable energy remains uneven across regions, particularly in Africa, where structural and macroeconomic constraints continue to hinder the energy transition.

Africa possesses abundant renewable energy potential, especially in solar, wind, and hydropower resources. Nevertheless, the continent still faces significant energy access challenges and relies heavily on traditional biomass and fossil fuels. According to recent estimates, renewable energy deployment in Africa is constrained by limited financial capacity, weak infrastructure, and macroeconomic vulnerabilities [5, 6]. These challenges highlight the importance of understanding the determinants of renewable energy consumption in African economies, particularly the role of macroeconomic, fiscal, financial, and structural factors.

Despite growing interest in renewable energy determinants, the empirical literature remains fragmented in several important ways. First, many studies focus on individual determinants such as economic growth, financial development, or public debt separately, without considering their combined and potentially interactive effects [7–9]. Second, existing research often assumes homogeneous effects across countries, overlooking the structural heterogeneity that characterizes African economies. However, countries differ substantially in terms of institutional quality, financial development, fiscal capacity, and natural resource endowment, which may lead to different energy transition trajectories. Third, most empirical analyses rely on conventional econometric approaches and do not fully exploit recent advances in machine learning methods capable of capturing nonlinear relationships and complex interactions among variables.

These limitations leave several important research questions unresolved. It remains unclear whether macroeconomic, fiscal, and structural factors influence renewable energy consumption uniformly across African countries, or whether distinct groups of countries follow different transition paths. Furthermore, the combined use of machine learning techniques and dynamic econometric models to identify heterogeneous patterns and long-run relationships remains largely unexplored in the African context.

To address these gaps, this study examines the combined effects of public debt, financial development, and key macroeconomic indicators—including gross domestic product (GDP), trade openness, and urbanization—on renewable energy consumption in African countries.

From a theoretical perspective, the relationship between macroeconomic, fiscal, financial, and environmental factors and renewable energy consumption can be explained through several transmission channels. First, economic growth influences renewable energy consumption through the income and investment channels. Higher GDP increases energy demand, government revenues, and private investment capacity, which may facilitate the adoption of renewable energy technologies and support the energy transition. Second, public debt affects renewable energy through the fiscal space and public investment channel. Moderate levels of public debt may finance renewable energy infrastructure and green innovation, while excessive debt may crowd out public investment due to increasing debt servicing obligations. Third, financial development influences renewable energy through improved access to credit, lower cost of capital, and risk diversification, which are essential for financing capital-intensive renewable energy projects. Finally, environmental and climatic conditions determine the natural potential for renewable energy production, such as solar and wind energy. Therefore, renewable energy consumption results from the interaction between economic capacity, fiscal policy, financial development, and environmental constraints.

Using panel data for 25 African economies and an integrated analytical framework that combines machine learning and econometric techniques, this study makes three main contributions to the literature. First, it identifies heterogeneous energy transition trajectories across African countries rather than assuming a common pattern. Second, it integrates macroeconomic, fiscal, financial, and environmental determinants within a unified analytical framework. Third, it introduces a hybrid methodology combining machine learning and Panel Autoregressive Distributed Lag (Panel ARDL) modeling, thereby enhancing robustness and providing more policy-relevant insights.

The remainder of the paper is organized as follows: Section 2 reviews the relevant literature, highlighting key findings and theoretical frameworks. Section 3 describes the data sources and methodology used in the analysis. Section 4 presents the results alongside their corresponding discussions. Finally, Section 5 concludes the paper with summary insights and policy recommendations.

2 Literature Review

To better understand the determinants of renewable energy development, the existing literature can be organized into several strands. The first strand examines the relationship between public debt and renewable energy, the second focuses on financial development and renewable energy, while a third strand considers the role of macroeconomic and structural factors. This thematic organization helps clarify the main findings, differences, and gaps in literature.

2.1 Public Debt and Renewable Energy

One of the major challenges confronting the global economy in the 21st century is the transition from a carbon-intensive energy model to a sustainable and environmentally friendly system. Achieving sustainable development goals requires the large-scale deployment of renewable energy sources, which provide environmental benefits while reducing dependence on imported energy and supporting employment creation [9, 10]. However, the development of renewable energy requires substantial financial resources, particularly due to the high initial investment costs associated with clean technologies.

In this context, public debt has emerged as a potential financing mechanism for supporting renewable energy deployment. Borrowed funds can be used to finance research and development, infrastructure investment, and policy incentives that encourage the adoption of renewable technologies [11]. Moreover, expansionary fiscal policies financed by debt may stimulate economic growth, thereby increasing demand for modern energy services and creating favourable conditions for renewable energy investment. However, the contribution of public debt is not unlimited.

Excessive debt levels may reduce fiscal space, as governments must allocate increasing resources to debt servicing, potentially crowding out green investments [12, 13].

Empirical literature provides mixed evidence regarding the impact of public debt on renewable energy consumption; however, several studies report a positive relationship. For instance, Przychodzen and Przychodzen [14] showed that public debt promotes renewable energy production in 27 transition economies over the period 1990–2014. Similarly, Florea et al. [15] found a positive association between public debt and renewable energy consumption in 11 emerging European Union (EU) economies between 1990 and 2016. In the same vein, Farooq et al. [11] demonstrated that public debt positively affects renewable energy utilization in a panel of 17 Organisation for Economic Co-operation and Development (OECD) countries spanning 1980–2020. Focusing on developing regions, Ashour and Sayed [16] reported that reductions in public debt are associated with lower levels of renewable energy consumption in 21 African and MENA countries from 1999 to 2021, which indirectly supports the positive role of public debt in financing renewable energy. More recently, Çetin [17] found that public debt significantly influences renewable energy supply in Central and Eastern European (CEE) countries over the period 1996–2019. Overall, these findings suggest that public borrowing can support renewable energy development, particularly when debt is used to finance public investment in energy infrastructure and green projects.

Conversely, other studies report a negative effect of public debt on renewable energy development. For example, Bese et al. [18] indicated that external debt contributed to increased emissions in China between 1978 and 2014, thereby indirectly limiting renewable energy deployment. Similarly, Wang et al. [19] documented an adverse impact of public debt on renewable energy use in BRICS countries over the period 1990–2016. This negative relationship is further confirmed by Hashemizadeh et al. [20] for 20 emerging economies. At the country level, Jabari et al. [21] found that external debt had a detrimental effect on renewable energy consumption in Turkey between 1980 and 2016. Evidence from developing regions also supports this view, as study [22] showed that rising public debt reduces renewable energy consumption in Sub-Saharan Africa, while Qamruzzaman et al. [23] reported similar results for 25 Sub-Saharan African countries over the period 2000–2021. Even in advanced economies, high debt levels may constrain green investment, as demonstrated by Auteri et al. [12] for G7 countries. Overall, these studies suggest that high public debt may crowd out public investment and limit financial resources available for renewable energy projects.

In addition to these contrasting findings, some studies identify non-linear relationships. Study [24] analyzed 71 developing countries between 2000 and 2020 and highlight a U-shaped effect. This shows that public debt promotes investment in renewable energy up to a certain threshold. Beyond this threshold, however, excessive debt becomes counterproductive and slows down further investment.

Overall, the literature suggests that the relationship between public debt and renewable energy is ambiguous. Moderate debt may facilitate financing for green investments, but excessive debt can hinder the energy transition. These mixed findings highlight the need for further investigation, particularly in regions with severe fiscal constraints [25–27].

2.2 Financial Development and Renewable Energy Consumption

The transition to renewable energy also depends heavily on the availability of financial resources. Renewable energy projects are capital-intensive and involve significant upfront costs, which may discourage investment in countries with underdeveloped financial systems [20, 28]. A well-developed financial sector can facilitate access to credit, improve capital allocation, enhance risk-sharing mechanisms, and support innovation in green technologies [29].

A first strand of the literature highlights the positive role of financial development in promoting renewable energy. Early evidence by Al-Mulali et al. [30] showed that financial depth contributes to environmental sustainability. Similarly, Thai-Ha et al. [7] found that financial development supports renewable energy capacity expansion. Using a panel data approach, Anton and Afloarei Nucu [31] revealed that financial development exerts a positive effect on renewable energy consumption across 28 EU countries over the period 1990–2015. In the same vein, Shahbaz et al. [8] demonstrated that financial development stimulates renewable energy demand, while Prempeh [32] confirmed this positive effect in the case of Ghana. Overall, these studies suggest that well-developed financial systems facilitate the mobilisation of financial resources required to support investment in renewable energy and the energy transition.

However, a second strand of the literature reports contrasting results. Some studies argue that financial development may increase overall energy demand and reinforce reliance on fossil fuels rather than renewable energy. For instance, Sadorsky [33] showed that financial development stimulates economic growth, which in turn increases energy consumption and carbon emissions. Similarly, Acheampong et al. [34] and Yang et al. [35] documented a positive relationship between financial development and CO₂ emissions, suggesting that financial expansion may support carbon-intensive industries rather than clean energy investment. In the case of Tunisia, Saadaoui and Chtourou [36] found that financial development reduces renewable energy consumption despite improvements in institutional quality, highlighting the importance of regulatory and governance frameworks in directing financial resources towards renewable energy projects. Overall, these findings suggest that financial development does not automatically promote renewable energy; its impact depends on the structure of the economy, energy policies, and the quality of institutions

and regulations.

Recent studies highlight that the impact of financial development on renewable energy consumption is not uniform and depends on the context. For instance, Alshagri et al. [37] found that while financial development positively affects renewable energy consumption in advanced economies, it negatively impacts emerging and developing countries. Similarly, Horky and Fidrmuc [9] reported a dichotomy: traditional financial institutions and banks favor carbon-intensive energy, negatively affecting renewable energy consumption, whereas developed capital markets have a positive influence, particularly in EU countries. These findings suggest that the relationship between financial development and renewable energy is complex and channel-dependent, highlighting the need to consider country characteristics and financial structures in empirical analyses.

In summary, the relationship between financial development and renewable energy remains inconclusive, as financial expansion may either promote green investments or increase fossil fuel dependence.

2.3 Macroeconomic and Structural Determinants

Beyond fiscal and financial factors, macroeconomic variables also play a key role in shaping renewable energy demand. Economic growth, trade openness, and urbanisation are frequently identified as important determinants. Rapid economic growth often increases energy demand and environmental pressure; however, higher income levels may also encourage the adoption of cleaner technologies, consistent with the environmental Kuznets curve hypothesis [24, 38].

Trade openness influences renewable energy consumption through increased production and international trade in energy-intensive goods [39]. Urbanisation also affects energy demand by increasing consumption while creating opportunities for economies of scale and renewable energy deployment [40, 41]. Empirical studies confirm these relationships. Akintande et al. [42] highlighted the role of urbanisation in promoting renewable energy demand, while Hashemizadeh et al. [20] found significant correlations between trade openness, urbanisation, and renewable energy consumption. Similarly, Khuong et al. [43] showed that economic growth significantly influences renewable energy use.

Overall, macroeconomic and structural factors play an important role in renewable energy deployment, although their effects vary across countries and levels of development.

2.4 Research Gap and Contribution

The literature reveals that public debt and financial development are important determinants of renewable energy, yet they are generally examined separately. The joint effect of these two factors remains largely unexplored, particularly in African economies, where high debt levels and limited financial depth may constrain the financing of renewable energy projects.

To address this gap, this study simultaneously examines public debt and financial development using a machine learning approach. This framework allows us to capture complex and potentially non-linear interactions between fiscal and financial constraints, thereby providing new insights into the drivers of renewable energy development in Africa.

3 Data and Methodology

The study constructs a panel dataset for twenty-five selected African countries covering the period 1985–2023. Annual data were collected from two main sources. Data on renewable energy consumption, financial development, urbanisation, and trade openness were obtained from the World Bank's World Development Indicators (WDI). Data on the public debt-to-GDP ratio and the real GDP growth rate were sourced from the International Monetary Fund (IMF). The choice of the study period and the countries included depends on the availability of data for the selected variables of interest.

The selected countries for the study are as follows: Burundi, Cameroon, the Central African Republic, Côte d'Ivoire, Gabon, Guinea, Guinea-Bissau, Kenya, Madagascar, Mali, Mozambique, Niger, Tanzania, Algeria, Botswana, Cape Verde, Egypt, Senegal, the Seychelles, South Africa, Tunisia, Congo, Gambia, Ghana and Sudan.

Renewable energy consumption refers to the proportion of total final energy consumption derived from renewable sources, such as hydro, solar, wind, geothermal, and biomass. It is expressed as a percentage of total final energy consumption and serves as an indicator of the contribution of renewable energy to a country's overall energy mix. In this study, renewable energy consumption is considered the dependent variable.

In the empirical energy economics literature, the share of renewable energy consumption is widely used to assess the role of renewables in the national energy mix and to facilitate cross-country and temporal comparisons of renewable energy adoption [44, 45]. We acknowledge that this indicator reflects the relative share of renewable energy rather than its absolute quantity. In rapidly growing economies, the percentage share may decline even if the absolute amount of renewable energy increases, due to faster expansion of total energy demand. This measurement issue is important for interpretation; results should therefore be understood as capturing changes in the proportion of energy derived from renewable sources within the overall energy consumption. Nevertheless, the share of renewable

energy remains a standard empirical measure in panel studies of renewable energy dynamics and policy analysis, as it explicitly reflects the degree of renewable energy adoption relative to total energy use [45].

This definition provides the basis for examining how various economic, financial, and policy factors influence renewable energy consumption across the selected African countries. The independent variables selected based on existing literature are presented below.

The public debt-to-GDP ratio is defined as the total public debt of a country expressed as a percentage of its GDP. Theoretically, public debt can play an important role in promoting the deployment of renewable energy, as the high costs of these energy sources often lead governments to borrow to finance their development. However, excessive debt may hinder investments in renewable energy.

Financial development, proxied by domestic credit to the private sector (% of GDP), is expected to have ambiguous effects on renewable energy consumption. On one hand, a more developed financial system can facilitate investments in renewable energy by improving access to credit and lowering financing costs, which would support higher consumption of renewable energy. On the other hand, if financial resources are primarily allocated to non-renewable sectors or used inefficiently, financial development may hinder the expansion of renewable energy, resulting in a negative effect.

It is acknowledged that domestic credit to the private sector is a relatively narrow proxy for financial development and may not fully capture the depth of the financial system or its capacity to fund long-term infrastructure projects. However, this measure was chosen because it is consistently available across all selected countries and years, allowing for a reliable panel analysis while avoiding selection bias. Furthermore, this indicator is widely used in empirical studies [46, 47] as it reflects the ability of the banking sector to provide necessary liquidity to private innovators, providing a comparable indicator of financial development across countries and over time.

Trade openness, measured as the sum of exports and imports relative to GDP, reflects a country's integration into the global trading system. Its effect on renewable energy consumption can be either positive or negative: greater openness may promote renewable energy by facilitating technology transfer and access to clean energy products, but it may also increase reliance on imported conventional energy sources, potentially reducing the share of renewables in the energy mix.

Real GDP growth rate is the most widely used measure of economic growth, following the United Nations' recommendation, as it reflects the overall expansion of a country's economic activity. Consequently, it is considered a key factor influencing renewable energy consumption, since higher economic growth can increase energy demand and create opportunities for investment in renewable energy projects.

Urban population, measured as the percentage of total population, may have a positive or negative effect on renewable energy consumption. While urbanization can facilitate access to renewable energy and modern infrastructure, it may also increase reliance on conventional energy sources.

Table 1 presents the descriptive statistics of the main variables used in empirical analysis. These variables include renewable energy consumption (REC), urbanisation rate (Urban), trade openness (Trade), GDP, public debt (DEBT), and financial development (FS). All variables have been standardised prior to analysis to ensure comparability across countries and to avoid scale-related biases in the modelling process.

Table 1. Descriptive statistics of key indicators

Statistic	REC	Urban	Trade	GDP	DEBT	FS
Mean	-1.187	0.658	0.467	-0.308	-0.047	0.817
Standard Deviation	0.765	0.598	1.203	0.194	0.834	1.378
Minimum	-2.084	-0.932	-1.012	-0.465	-1.629	-0.778
Maximum	-4.263×10^{-17}	1.875	4.971	0.497	3.349	5.468
Jarque-Bera test	33.548	11.751	272.934	148.733	156.150	145.809
Prob.	0.000	0.000	0.000	0.000	0.000	0.000

Note: REC, renewable energy consumption; Urban, urbanisation rate; Trade, trade openness; GDP, gross domestic product; DEBT, public debt; FS, financial development.

The descriptive statistics reveal a diverse set of properties across the variables. Nonetheless, examining the levels and dispersion of these variables is crucial to understand their distributional characteristics. The dataset exhibits a range of distinct properties across the variables. Nonetheless, the Jarque-Bera test holds significant importance as it evaluates the normality of a given dataset. The evaluation assesses the adherence of the data to a normal distribution, which carries various implications contingent upon the test outcomes. These properties justify the need for advanced analytical methods capable of capturing heterogeneity and dynamic behavior across countries and over time.

Given this structural diversity, our approach involves initial segmentation through clustering techniques to group countries with similar profiles. This reduces the complexity associated with heterogeneity and allows for tailored analysis within each cluster, thus enhancing the relevance of the subsequent dynamic modeling with the Panel ARDL approach. The methodology includes handling missing data using an iterative imputer, excluding countries with more

than 20% missing values to ensure data reliability. The dataset is then split into training (70%), validation (15%), and test (15%) sets, while preserving the temporal structure to prevent data leakage, ensuring robustness and consistency in the analysis.

First, we implement the Light Gradient Boosting Machine (LightGBM) model [48], a gradient boosting framework based on decision trees, to identify the main drivers of renewable energy consumption. Recent benchmarking studies show that this approach performs competitively relative to alternatives such as eXtreme Gradient Boosting (XGBoost) and Categorical Boosting (CatBoost) [49]. The model minimises the mean squared error (MSE) loss function:

$$L_{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

At each iteration, the model is updated using first- and second-order gradients:

$$L^{(t)} \approx \sum_{i=1}^n \left[l(y_i, F^{(t-1)}(x_i)) + g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2 \right] + \Omega(f_t) \quad (2)$$

Hyperparameters are tuned using Bayesian optimisation via Optuna, combined with GroupKFold cross-validation ($k = 5$) to account for the panel structure and avoid information leakage across countries. The main parameters include the number of trees (typically 100–500), learning rate (0.01–0.1), maximum tree depth (3–10), and number of leaves. Regularisation parameters (λ, γ) are also optimised to prevent overfitting and enhance generalisation. Feature importance is computed based on cumulative gain:

$$\text{Gain} = \frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right) - \gamma \quad (3)$$

The second step involves temporal segmentation of countries using the Dynamic Time Warping (DTW) distance, which aligns time series with different temporal dynamics:

$$\text{DTW}(x, y) = \min_{\pi \in A(x, y)} \sum_{(i, j) \in \pi} d(x_i, y_j) \quad (4)$$

The cost matrix is constructed recursively.

$$D(i, j) = d(x_i, y_j) + \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\} \quad (5)$$

where, i and j are the indices of the time points in the two sequences, and $D(i, j)$ is the cumulative distance at position (i, j) in the cost matrix.

Prior to clustering, time series are normalised (z-score standardisation) to ensure comparability across countries. A constraint window (Sakoe-Chiba band) is applied to limit excessive warping and improve computational efficiency. The DTW distance matrix is then used in conjunction with a k-means clustering algorithm, which is more robust to noise and outliers than k-means when using non-Euclidean distances. The optimal number of clusters is determined by using internal validation metrics such as the silhouette score and the elbow method. This step identifies homogeneous groups of countries with similar renewable energy adoption trajectories [50].

The third step involves validating the robustness of the variable importance results obtained from LightGBM using Long Short-Term Memory (LSTM) networks [51]. For this purpose, input data are transformed into sequences via a sliding window approach of length T . The LSTM architecture incorporates one to three hidden layers, each containing between 32 and 128 units, and uses tanh and sigmoid activation functions. Dropout regularisation (typically set to between 0.2 and 0.5) is then applied to reduce overfitting and improve model generalisation.

The model is trained using the Adam optimiser with a learning rate of 0.001, a batch size of 32 and between 50 and 100 epochs. Early stopping based on validation loss is used to prevent overfitting. The LSTM computes hidden states as follows:

$$h_t = \text{LSTM}(x_t, h_{t-1}) \quad (6)$$

where, t denotes the time step, x_t is the input vector at time t , and h_t is the hidden state at time t .

Once the model has been trained, the relative importance of the input variables, as derived from the LSTM (e.g. via input perturbation or gradient-based methods), is compared with the importance rankings produced by LightGBM. Consistency between the two rankings, as determined by metrics such as root mean squared error (RMSE) and mean absolute error (MAE) on the test set, indicates robustness. Rather than aiming to forecast future renewable energy trajectories, this approach serves as a cross-validation tool to confirm that the key determinants identified by LightGBM are not artefacts of the chosen method.

4 Results and Discussion

The Results section is divided into subsections. It should detect important variables and attempt to cluster African countries.

4.1 LightGBM Variable Importance

A data standardisation step was systematically carried out before applying the LightGBM and DWT methods. This step involved normalising all explanatory and endogenous variables to ensure comparability by removing effects related to differences in scale, units of measurement, and statistical distributions. Following this pre-processing step, the LightGBM method was used to evaluate the importance of the variables and develop high-performance predictive models. Figure 1 shows how the LightGBM model performed, illustrating the quality of the prediction obtained.

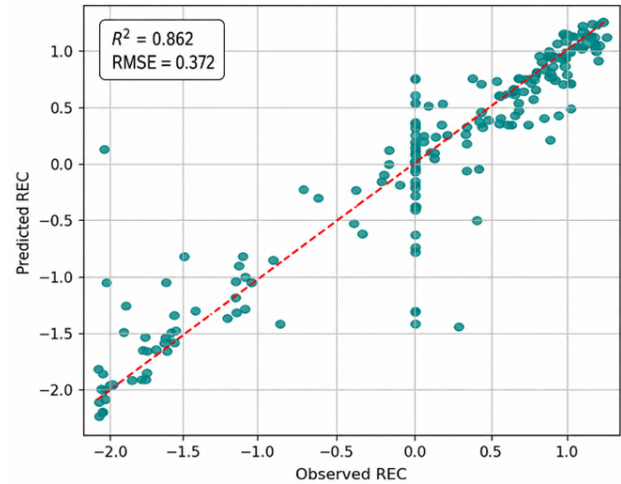


Figure 1. Real vs. predicted renewable energy consumption using Light Gradient Boosting Machine (LightGBM)

Figure 1 shows how well the LightGBM model predicts renewable energy consumption in our sample. The quality of the fit observed demonstrates the robustness of the model when analysing the dynamics of renewable energy consumption. To gain a deeper understanding of the mechanisms underlying this predictive effectiveness, it is useful to examine the relative contribution of the various explanatory variables included in the model. Analysing the importance of these factors provides insight into the structural determinants that underlie the model’s ability to explain energy variations and dynamics at a national level. Figure 2 presents an assessment of the importance of the variables.

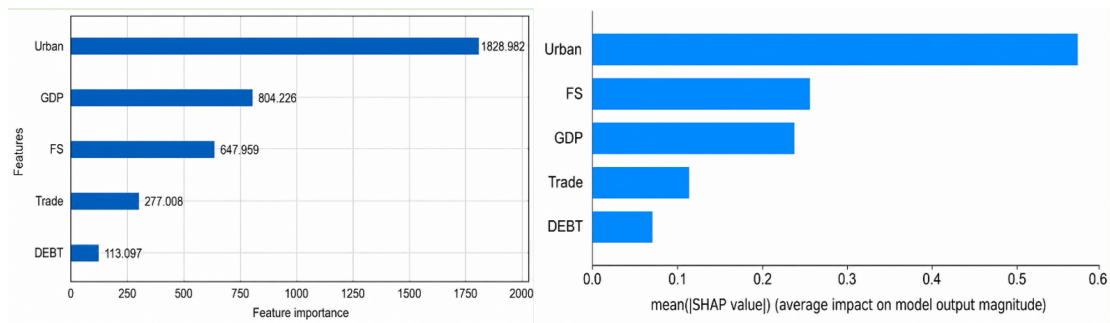


Figure 2. Relative importance and impact (SHapley Additive exPlanations) of explanatory variables in Light Gradient Boosting Machine (LightGBM) model for renewable energy consumption

Note: Urban, urbanisation rate; Trade, trade openness; GDP, gross domestic product; DEBT, public debt; FS, financial development.

Figure 2 presents the SHapley Additive exPlanations (SHAP)-based importance of the explanatory variables in the LightGBM model, providing both a ranking and a quantitative assessment of their contribution to renewable energy consumption in the African panel. The results reveal a highly uneven distribution of influence across variables, highlighting the dominance of structural factors.

Urbanisation (Urban) emerges as the most influential determinant by a wide margin, with an importance score of 1828.982 and an average SHAP contribution exceeding 0.55. This value is more than twice that of the second most important variable, confirming the central role of urban dynamics in shaping renewable energy demand. From an

economic standpoint, this finding reflects the fact that urban expansion in Africa is closely associated with increased energy needs, improved infrastructure, and greater access to modern and decentralized renewable technologies. Urban areas also benefit from economies of scale and more efficient energy distribution systems, which facilitate the integration of renewables into the energy mix.

Financial development (FS) and GDP follow as the second and third most important variables, with SHAP importance scores of 647.959 and 804.226, respectively, and average contributions around 0.23–0.25. Their relatively close magnitudes suggest complementary roles in driving the energy transition. Financial development enhances access to capital, reduces financing constraints, and supports investment in renewable energy projects, while higher GDP reflects stronger economic capacity and increased public and private investment in clean energy infrastructure. The positive contribution of these variables indicates that improvements in economic and financial conditions act as key enablers of renewable energy adoption.

In contrast, trade openness exhibits a more moderate influence, with an importance score of 277.008 and a mean SHAP value close to 0.11. This suggests that while trade can facilitate the diffusion of renewable technologies through imports of capital goods and knowledge transfer, its direct impact remains limited compared to domestic structural factors.

Public debt (DEBT) appears as the least influential variable, with a relatively low importance score of 113.097 and a mean SHAP contribution below 0.08. This weak contribution indicates that the effect of debt on renewable energy consumption is likely indirect and context dependent. While high debt levels may constrain fiscal space for green investments, the absence of a strong direct effect suggests that other structural and institutional factors play a more decisive role.

Overall, the SHAP analysis goes beyond a simple ranking by quantifying the magnitude of each determinant and highlighting the economic mechanisms at play. The findings clearly indicate that urbanisation, together with economic and financial development, constitutes the primary driver of renewable energy consumption in Africa, whereas external and fiscal variables exert a more limited and indirect influence.

4.2 Dynamic Time Warping-Based Clustering of African Countries

A thorough analysis of intra-cluster inertia enables the quality of the segmentation to be assessed. Figure 3 illustrates this evolution in relation to the number of clusters (k), calculated from the DTW distance applied to renewable energy consumption trajectories. The analysis reveals a rapid decrease in inertia when k increases from 2 to 3, followed by a more gradual reduction for values greater than three. This suggests that adding more clusters only marginally improves intra-cluster cohesion. Consequently, segmentation into three groups appears to be the optimal solution for classifying the African countries studied.

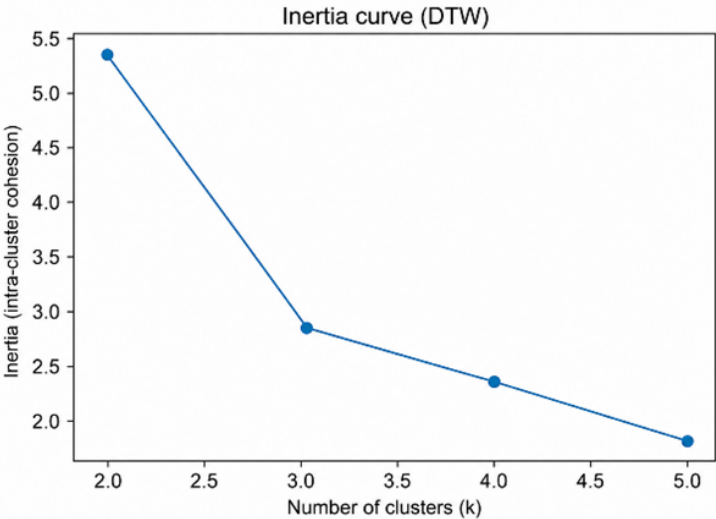


Figure 3. Optimising the number of clusters (DTW method)

Note: DTW, Dynamic Time Warping.

Adopting this number of clusters ($k = 3$) enables the classification of the 25 African countries into three distinct groups, making it possible to identify different energy transition dynamics according to their temporal trajectories. Figure 4 shows how the countries are distributed across the clusters.

Figure 4 shows a heatmap illustrating the DTW distance matrix which was calculated using the temporal trajectories of renewable energy consumption in a sample of 25 African countries. Each cell of the matrix represents the distance

between two countries' time series: lower values (represented by darker shades) indicate a high degree of similarity in the evolution of renewable energy consumption, while higher values (lighter shades) reflect greater divergence in dynamic patterns. By construction, the diagonal elements are equal to zero, corresponding to each country's consumption pattern compared to itself.

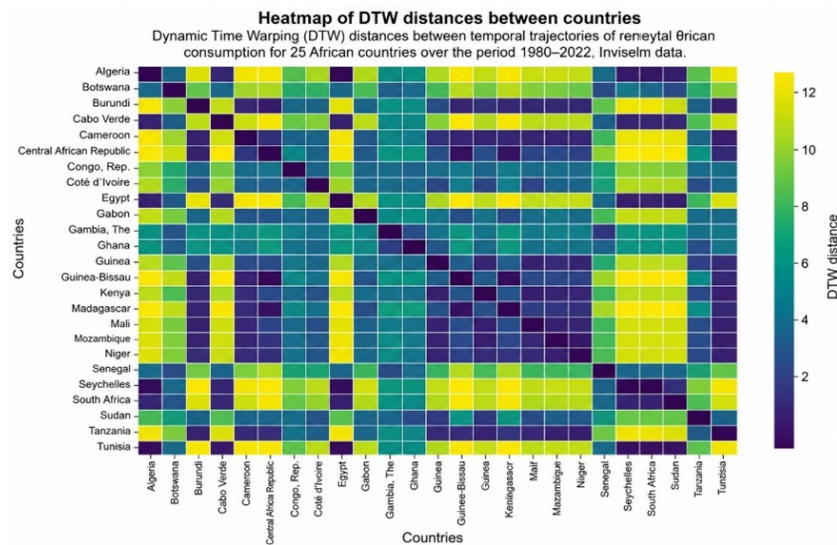


Figure 4. Heatmap of temporal distances (DTW) between African countries (renewable energy consumption)

Note: DTW, Dynamic Time Warping.

To facilitate interpretation, the colour scale captures the full distribution of DTW distances, enabling readers to visually assess both local similarities (between individual countries) and global structures (between groups of countries). Contiguous dark-coloured blocks along the diagonal indicate groups of countries with similar temporal dynamics, while lighter off-diagonal regions highlight differences between these groups.

This visualisation reveals a clear block structure, suggesting the presence of homogeneous clusters. These patterns are not random but reflect underlying similarities in energy transition paths. These are consistent with the results obtained from the temporal clustering algorithm. Interpreting the heat map alongside the clustering procedure strengthens the robustness of the classification and provides intuitive validation of the clustering structure.

A combined analysis of the heat map and classification identifies three distinct groups:

Cluster 1 includes 13 countries: (Burundi, Cameroon, the Central African Republic, Côte d'Ivoire, Gabon, Guinea, Guinea-Bissau, Kenya, Madagascar, Mali, Mozambique, Niger and Tanzania). These countries have converging renewable energy consumption trajectories, characterised by low DTW distances within the group and represented by dark squares around the sub-diagonal of the matrix. They have converging renewable energy consumption trajectories, characterised by low DTW distances within the group and it can be interpreted as a group of low-capacity states with limited fiscal and institutional resources. These countries tend to rely heavily on traditional biomass and small-scale renewable sources, due to structural constraints such as weak industrialization, limited access to fossil fuel infrastructure, and constrained public investment capacity. Politically, energy transitions in this cluster are often passive rather than strategic, shaped more by necessity than by long-term planning.

Cluster 2 includes eight countries (Algeria, Botswana, Cape Verde, Egypt, Senegal, the Seychelles, South Africa and Tunisia) that share similar energy trajectories. They are more distant from the other groups and reflect a particular energy dynamic, shaped by their specific national and regional contexts. These countries are characterized by more organized and policy-oriented energy transitions, often underpinned by stronger government capacity, diversified economies, and more clearly defined national energy strategies. Politically, this group reflects systems in which governments play an active role in shaping energy policy, aiming to balance reliance on fossil fuels with the expansion of renewable energy. In some cases, particularly in resource-rich countries, energy policy is also linked to broader issues of economic diversification and political stability., shaped by their specific national and regional contexts.

Cluster 3 comprises four countries: Congo, Gambia, Ghana and Sudan. This group is also internally cohesive, as can be seen from its greater proximity on the heat map. However, the brighter colours highlight notable differences compared to the other clusters. Historically, these countries have been characterised by recurring political instability and frequent regime changes. Sudan experienced military coups in 1989 and 2019, while the Republic of the Congo and The Gambia underwent democratic transitions in 1991 and 2016, respectively. Although Ghana is relatively more stable, it remains vulnerable to economic shocks. Thus, this cluster characterises energy systems in transition

or undergoing adjustment, where energy policy remains unstable and sensitive to external shocks such as political changes, economic crises and fluctuations in international support.

These structural differences in energy trajectories reflect disparities in the countries' socio-economic contexts. To better understand these dynamics, the specific economic characteristics of each group must be analysed. Studying the average economic profiles of each cluster, excluding the renewable energy component, sheds light on the potential links between energy morphology and macroeconomic indicators. The aim of this step is to identify the relationships between cluster structuring and economic dimensions such as GDP, debt and level of urbanisation, to deepen our understanding of the factors influencing renewable energy consumption in Africa. Figure 5 summarises this analysis by providing a clear representation of these estimates.

Figure 5 presents a heatmap summarizing the average standardized economic characteristics of the three identified clusters, based on key determinants of renewable energy consumption, namely urbanisation, trade openness, GDP growth, public debt, and financial development. The reported values correspond to normalized averages within each cluster, where positive values indicate above-average levels relative to the full sample, and negative values indicate below-average levels. The colour gradient facilitates interpretation, with warmer tones (red) reflecting higher values and cooler tones (blue) indicating lower values.

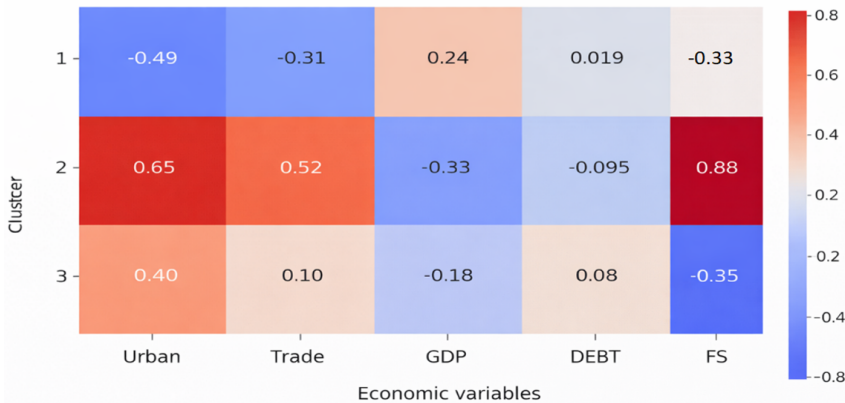


Figure 5. Average standardized economic profiles by cluster

Note: Urban, urbanisation rate; Trade, trade openness; GDP, gross domestic product; DEBT, public debt; FS, financial development.

Cluster 1 is characterized by below-average levels of urbanisation (-0.49) and trade openness (-0.31), combined with slightly above-average GDP performance (0.24) and relatively neutral debt levels. This configuration suggests the presence of economies where growth dynamics are not yet supported by strong structural transformation, particularly in terms of integration into global markets and urban development.

Cluster 2 exhibits a markedly different profile, with high levels of urbanisation (0.65), trade openness (0.52), and especially financial development (0.88), the highest among all clusters. However, GDP growth appears below average (-0.33), which may reflect structural imbalances or a reliance on sectors with limited productivity gains. This group represents relatively advanced and integrated economies but potentially exposed to macroeconomic vulnerabilities.

Cluster 3 displays intermediate urbanisation (0.40) and moderate trade integration (0.10) but is distinguished by negative GDP performance (-0.18) and relatively higher public debt (0.08). Financial development is also below average (-0.35), suggesting structural and financial constraints. These characteristics indicate that fiscal pressure and limited financial depth may hinder the capacity of these countries to support sustained energy transitions.

Overall, this typology highlights the strong heterogeneity in the structural drivers of renewable energy consumption across African countries. It shows that renewable energy consumption dynamics are not solely driven by economic growth but are critically shaped by broader structural factors such as urbanisation, trade integration, and financial system development, with effects that vary significantly across clusters.

Having clustered the 25 African countries using the DTW approach, which groups countries according to the similarity of their temporal energy and economic dynamics and following a preliminary identification of key determinants of renewable energy consumption, this study proceeds to a more in-depth econometric analysis at the cluster level. Specifically, this study estimates the Pooled Mean Group Autoregressive Distributed Lag (P-ARDL) model separately for each cluster, capturing both long-run relationships and short-run adjustment dynamics. This stepwise approach enables us to account for structural heterogeneity across clusters while retaining the potential for shared long-term equilibrium relationships within each group. Combining clustering and econometric modelling provides a more nuanced understanding of how drivers of renewable energy consumption differ across distinct country profiles, reflecting their underlying economic and institutional characteristics.

However, prior verification of the stationarity properties of the series is required for the implementation of this

cluster modelling. Unit root tests, summarised in Table 2 and applied to each group, reveal significant heterogeneity in the order of integration of the variables. Some are stationary at level [I(0)], while others are only stationary after first differentiation [I(1)]. Given these characteristics, the ARDL approach is particularly appropriate in this context, as it enables I(0) and I(1) series to be analysed together, robust cointegration relationships to be estimated and the short-term adjustment mechanisms specific to each cluster to be examined.

Table 2. Unit roots tests for cluster 1, 2 and 3

Variables		Levin, Lin & Chu t^*	ADF-Fisher Chi-Square	PP-Fisher Chi-Square
Cluster 1				
REC	Level	-2.79 (0.002)	26.34 (0.444)	26.97 (0.411)
	1st diff	-21.21 (0.000)	413.70 (0.000)	413.90 (0.000)
Urban	Level	-1.73 (0.042)	44.61 (0.013)	278.50 (0.000)
GDP	Level	-3.97 (0.000)	52.34 (0.000)	86.40 (0.000)
FS	Level	-2.89 (0.002)	44.03 (0.015)	39.70 (0.041)
Trade	Level	-3.44 (0.000)	40.91 (0.031)	42.90 (0.019)
DEBT	Level	-3.36 (0.000)	40.36 (0.036)	44.90 (0.012)
Cluster 2				
REC	Level	-2.25 (0.012)	16.53 (0.416)	16.70 (0.403)
	1st diff	-16.70 (0.000)	258.90 (0.000)	282.20 (0.000)
Urban	Level	3.45 (0.999)	34.39 (0.004)	10.90 (0.812)
	1st diff	-2.47 (0.006)	37.16 (0.002)	41.30 (0.000)
GDP	Level	-5.42 (0.000)	101.05 (0.000)	345.50 (0.000)
FS	Level	-1.49 (0.067)	20.70 (0.188)	20.10 (0.214)
	1st diff	-14.90 (0.000)	242.10 (0.000)	242.60 (0.000)
Trade	Level	-2.20 (0.013)	22.90 (0.116)	21.19 (0.171)
	1st diff	-17.70 (0.000)	312.90 (0.000)	516.10 (0.000)
DEBT	Level	-1.62 (0.052)	13.20 (0.651)	18.60 (0.288)
	1st diff	-17.00 (0.000)	362.90 (0.000)	363.20 (0.000)
Cluster 3				
REC	Level	-2.30 (0.010)	14.86 (0.062)	15.60 (0.047)
	1st diff	-11.10 (0.000)	119.06 (0.000)	142.80 (0.000)
Urban	Level	-1.74 (0.040)	10.49 (0.232)	16.01 (0.042)
	1st diff	-0.86 (0.192)	15.10 (0.056)	14.20 (0.075)
GDP	Level	-3.04 (0.000)	45.10 (0.000)	54.50 (0.000)
FS	Level	-1.93 (0.026)	10.30 (0.244)	10.10 (0.255)
	1st diff	-12.71 (0.000)	164.40 (0.000)	164.8 (0.000)
Trade	Level	-0.97 (0.166)	17.16 (0.028)	15.90 (0.042)
	1st diff	-12.30 (0.000)	143.70 (0.000)	170.10 (0.000)
DEBT	Level	-3.44 (0.000)	21.60 (0.005)	23.60 (0.002)

Note: ADF, Augmented Dickey-Fuller; PP, Phillips-Perron; REC, renewable energy consumption; Urban, urbanisation rate; Trade, trade openness; GDP, gross domestic product; DEBT, public debt; FS, financial development.

The specification of our P-ARDL model is based on the following equation:

$$\begin{aligned}
 REC_{it} = & \alpha_i + \sum_{p=1}^P \beta_p REC_{i,t-p} + \sum_{j=1}^k \sum_{q=0}^{Q_i} \gamma_{jq} Urban_{j,i,t-q} + \sum_{j=1}^k \sum_{q=0}^{Q_i} \delta_{jq} GDP_{j,i,t-q} + \sum_{j=1}^k \sum_{q=0}^{Q_i} \gamma_{jq} FS_{j,i,t-q} \\
 & + \sum_{j=1}^k \sum_{q=0}^{Q_i} \gamma_{jq} Trade_{j,i,t-q} + \sum_{j=1}^k \sum_{q=0}^{Q_i} \gamma_{jq} DEBT_{j,i,t-q} + \varepsilon_{it}
 \end{aligned} \quad (7)$$

where, i and t index represent the country and time dimensions, respectively, p denotes the lag order for the dependent variable, j indexes the explanatory variables ($j = 1, \dots, k$), and q denotes the lag order for the explanatory variables.

The results are summarised below in Table 3.

Estimates from the Pooled Mean Group (PMG) error correction model, when applied to the three identified clusters, reveal the complex, multi-temporal dynamics that underlie the energy transition on the African continent. The PMG model was particularly relevant for this study as it assumes a homogeneous long-term relationship within each group of countries—a plausible assumption for structurally similar economies that are often subject to common

shocks—while allowing for complete heterogeneity in short-term dynamics and adjustment speeds. This approach thus captures the essence of Africa’s energy situation: a shared aspiration for long-term equilibrium, but with distinct transition trajectories and varying immediate influences.

The results, detailed in Table 3 and summarized by cluster, confirm significant convergence towards this long-term equilibrium, as evidenced by negative and statistically significant error correction coefficients. Nonetheless, this convergence is driven by contrasting long-term forces that sometimes promote, and other times strongly inhibit, progress depending on the structural characteristics and level of development of each group. While many short-term coefficients are statistically insignificant, this does not undermine the model’s validity; rather, it highlights the complexity and variability of immediate responses, where short-term fluctuations may be influenced by transient shocks or context-specific factors. The significance of the error correction coefficients across clusters reinforces the idea that, despite these short-term fluctuations, there exists a robust long-term convergence process.

Table 3. Long- and short-term Pooled Mean Group (PMG) estimation coefficients by cluster

Variables	Cluster 1: PMG(1,4,1,4,0,0)		Cluster 2: PMG(1,1,4,3,4,4)		Cluster 3: PMG(1,1,4,1,1,1)	
	Coefficient	<i>p</i> -Value	Coefficient	<i>p</i> -Value	Coefficient	<i>p</i> -Value
Long-Run (Pooled)						
Urban	0.014	0.664	0.418	0.024	-1.093	0.002
GDP	-0.214	0.000	1.723	0.002	3.045	0.010
FS	-0.429	0.000	0.148	0.419	-0.371	0.133
Trade	-0.031	0.598	-0.021	0.865	-0.104	0.154
DEBT	-0.527	0.000	-0.042	0.614	-0.110	0.000
Short-Run (Mean-Group) Coefficients						
COINTEQ	-0.300	0.000	-0.251	0.003	-0.247	0.080
D(Urban)	9.298	0.650	-0.625	0.964	17.110	0.204
D(Urban(-1))	-13.344	0.595	-176.717	0.131	—	—
D(Urban(-2))	-1.111	0.956	281.172	0.090	—	—
D(Urban(-3))	7.827	0.519	-107.834	0.060	—	—
D(GDP)	-6.140	0.151	-210.773	0.422	25.573	0.678
D(GDP(-1))	—	—	76.599	0.709	116.722	0.240
D(GDP(-2))	—	—	-118.360	0.456	-52.996	0.696
D(GDP(-3))	—	—	-774.149	0.261	267.880	0.012
D(FS)	-0.046	0.872	-0.041	0.887	-0.091	0.733
D(FS(-1))	0.224	0.044	-0.206	0.362	—	—
D(FS(-2))	-0.292	0.014	-0.354	0.024	—	—
D(FS(-3))	0.137	0.181	—	—	—	—
D(Trade)	—	—	0.095	0.608	0.039	0.576
D(Trade(-1))	—	—	0.339	0.005	—	—
D(Trade(-2))	—	—	-0.055	0.815	—	—
D(Trade(-3))	—	—	0.084	0.425	—	—
D(DEBT)	—	—	-0.004	0.976	-0.053	0.391
Diagnostics						
Log-likelihood	334.116		106.508		122.407	
Wald Test						
<i>F</i> -statistic	74.740	0.000	3.480	0.004	6.286	0.000
Chi-square	373.700	0.000	17.402	0.003	31.432	0.000

Note: Urban, urbanisation rate; Trade, trade openness; GDP, gross domestic product; DEBT, public debt; FS, financial development. A dash (–) denotes no data.

Overall, these findings suggest that policy efforts should account for both delayed and immediate effects, recognizing that short-term dynamics may be less predictable and more dependent on country-specific contexts. The dominant role of long-term forces underscores the importance of strategies that foster sustained structural adjustments, while acknowledging that diverse trajectories—shaped by factors such as urbanization, economic growth, and financial development—characterize the energy transition across Africa.

Cluster 1: An Energy Transition Under Constraint: The Burden of Debt and Economic Stagnation

For cluster 1, the negative and significant GDP coefficient suggests that these countries’ current growth trajectory is not decoupled from conventional energy sources. In line with the ‘infrastructure constraint’ theory, rapid economic growth, which is often driven by resource extraction and nascent industrialisation, is primarily fuelled by expanding fossil fuel capacities or traditional biomass. These are still perceived as cheaper and easier to deploy in the short term [52]. Our results are consistent with Jeetoo [53], who, using a Durbin spatial model across 41 Sub-Saharan African countries between 2002 and 2015, demonstrated that GDP per capita negatively affects renewable energy consumption, which decreases as income increases. This is primarily because richer countries favour fossil fuels. The

quality of governance, on the other hand, has a positive effect. These results also align with those of Nawaz and Rahman [54], who identified a 'U' relationship between GDP and renewable energy consumption. At low incomes, economic growth reduces consumption, but it rebounds to higher levels, with human capital playing a key role.

Consequently, while the absolute consumption of renewable energy may increase, its relative share of the total energy mix is declining—a classic rebound effect in the context of capacity constraints.

The strongly positive coefficient for the urban population is particularly striking. Accelerated urbanisation in Africa is creating an exponential demand for energy to power housing, transport and services. This demand is often met by centralised electricity networks that depend on thermal energy (fuel oil, gas and coal), as well as by the massive consumption of biomass (charcoal) in peri-urban areas. These findings are consistent with study [55], which documented that urbanisation in Sub-Saharan Africa has a significant and varying impact on overall energy consumption, with urban growth driving increased energy demand but exerting differing effects on electricity use per capita. Such dynamics support the 'carbon lock-in' hypothesis [56], whereby newly built urban infrastructure relies on non-renewable energy technologies, making future transitions more costly and complex.

The negative effect of financial development on the proportion of renewable energy consumed is counterintuitive. This suggests that the financial sector in these countries is not sufficiently oriented towards financing green projects. Instead, it tends to channel capital towards traditional or speculative activities, which are considered less risky. This configuration reflects a cautious financial structure where institutions prioritise supporting established economic sectors over the innovative and long-term investments required for the energy transition. These findings are consistent with previous studies [57, 58], which also highlight the limited role of financial development in promoting renewable energy in similar contexts.

The insignificant levels of public debt (DEBT) and trade openness (Trade) in these 13 countries suggest that these factors are not key determinants of the long-term share of renewable energy. This is due to fiscal margins already being constrained, which limit the impact of debt, and the fact that trade is largely dominated by the export of unprocessed raw materials, which have little impact on the domestic energy structure.

The coefficient of return for the long-term relationship (COINTEQ = -0.300) is highly significant. This fundamental result has several implications. It suggests that around 30% of the discrepancy between current renewable energy consumption and the long-term equilibrium level is resolved annually. In macroeconomic models, an adjustment rate of 30% is considered relatively fast. This suggests that, despite structural rigidities, the energy mix in these countries can adapt. There is an intrinsic capacity for adaptation. This speed presents a strategic opportunity: well-targeted public policies, such as subsidies and investments in networks, can produce significant effects in a relatively short period of time. However, it also indicates a certain volatility or low persistence of positive shocks in the absence of structural measures to anchor them.

Analysis of the heterogeneity of the short-term effects reveals that financial development generates a nonlinear and sequential dynamic characterised by a significant positive effect at first lag (coefficient = 0.224, $p < 0.05$), followed by a negative reversal at second lag (coefficient = -0.292, $p < 0.05$). This pattern may reflect an initial short-run boost in access to finance and investment in renewable energy, followed by an adjustment or partial reversal in the subsequent period. By contrast, economic growth and international trade are not statistically significant in the short run, suggesting that their effects on renewable energy consumption are mainly transmitted through longer-term channels. Overall, these findings point to an adjustment process rather than a stable short-run effect, highlighting the temporary and ambivalent role of financial development in supporting the energy transition.

Cluster 2: A transition driven by the internal dynamics of emerging African economies

The results of the PMG(1,1,4,3,4,4) model for Cluster 2 show that there are positive and significant determinants of renewable energy consumption. The GDP coefficient (1.723, $p = 0.002$) is notably high, suggesting that economic growth increases the proportion of renewable energy in the energy mix. This finding aligns with Ibrahim and Ibrahim [59], who reported that a 1% rise in renewable energy share is associated with a 0.15% increase in economic growth. Similarly, Agyeman and Lin [60] observed that in Tunisia, technological progress enhances output driven by renewable energy inputs, whereas in Algeria, Egypt, and Morocco, capital-biased technological change tends to suppress this relationship. Moreover, Khobai [61] found that in South Africa, renewable energy consumption Granger-causes long-term economic growth, with a bidirectional relationship between poverty and growth. This can be attributed to the proactive policies observed in leading countries such as South Africa, Egypt and Morocco, which have made significant investments in solar and wind power as part of their economic diversification strategies, thereby reducing their dependence on fossil fuels.

The effects of urbanisation on the energy transition are complex. In this context, our results demonstrate that urbanisation has a notably positive impact (coefficient = 0.418, $p = 0.024$), especially within more developed clusters. Here, urban modernisation contributes to the development of sustainable energy infrastructure and increases demand for green services. This contrasts with the 'carbon lock-in' phenomenon often observed in emerging economies at an early stage of development. Our findings align with those of Prempeh et al. [57], who emphasised that urbanisation could have a substantial detrimental impact on renewable energy consumption. However, study [42] showed that the

urban population is a key determinant of increased renewable energy consumption, suggesting that the impact varies depending on the context. These findings emphasise the importance of adopting a nuanced approach to urbanisation management to promote the transition to sustainable energy. Our results emphasise the importance of adopting a nuanced approach to urbanisation management to facilitate the transition towards more sustainable energy.

Financial and trade channels remain marginal; financial development (coefficient = 0.148, $p = 0.419$) and trade (coefficient = -0.021, $p = 0.865$) are not significant. This highlights that the transition relies mainly on internal dynamics rather than external technology transfers.

The error correction term (COINTEQ = -0.251, $p = 0.003$) confirms rapid convergence towards long-term equilibrium (with an annual adjustment of approximately 25%), as validated by the Wald test ($F = 3.480$, $p = 0.004$).

In summary, Cluster 2 embodies an endogenous energy transition driven by growth and virtuous urbanisation, in which financial levers play a secondary role.

However, this model has significant structural limitations. The lack of a significant impact of financial development (coefficient = 0.148, $p = 0.419$) reveals the inadequacy of domestic financing mechanisms. As Mazzucato and Semieniuk [62] have demonstrated, although African banking systems have been stabilised, they remain reluctant to finance renewable energy projects that are perceived as risky. This confirms the need to develop financial instruments that are adapted to local specificities.

The short-run coefficients also suggest a largely temporary and uneven pattern of adjustment. Trade openness has a positive and significant effect only at the first lag, suggesting that the benefits of trade integration are short-lived and do not persist across subsequent periods. This suggests that the benefits of trade may be realised quickly through improved access to renewable energy technologies, imported inputs or investment flows, but these effects are not sustained in later periods. The other short-run coefficients are mostly statistically insignificant, suggesting that their immediate impact on renewable energy consumption is limited. Overall, the results imply that short-run dynamics in this cluster are driven more by temporary adjustments than durable effects.

Cluster 3: A Conflicting Path Between the Desire for Transition and Structural Constraints

The results of the PMG(1,1,4,1,1,1) model for Cluster 3 reveal an energy trajectory characterised by significant tensions, with powerful driving forces being offset by substantial structural constraints. GDP has a significant positive effect of large magnitude (coefficient = 3.045, $p = 0.010$), suggesting that economic growth allocates substantial resources to renewable energies, potentially through ambitious public policies or favourable natural resources (e.g. solar or hydroelectric).

However, this momentum is offset by urbanisation, which has a negative and highly significant impact (coefficient = -1.093, $p = 0.002$). This illustrates a structural lock-in whereby rapid and often unplanned urbanisation generates an immediate energy demand that is primarily met by cheap fossil fuels, which is detrimental to long-term renewable investments. These findings are corroborated by Dingru et al. [63], who conducted a study at the sub-Saharan African level, and by Richmond et al. [64], who analysed 32 sub-Saharan African countries between 1990 and 2015. Both studies concluded that urbanisation 'hampers efforts to promote renewable energy consumption'. These studies emphasise the importance of adopting more sustainable urban strategies to prevent the reinforcement of this structural lock-in.

These constraints are reinforced by the macroeconomic environment: public debt (coefficient = -0.110, $p = 0.000$) weighs heavily, limiting budgetary capacity for green investment, while financial development (coefficient = -0.371, $p = 0.133$) and trade openness (-0.104, $p = 0.154$) remain insignificant, confirming the marginal role of financial and external channels.

The error correction term (COINTEQ = -0.247, $p = 0.080$) indicates convergence towards long-term equilibrium, albeit at the limit of significance (approximately 25% annual adjustment), which is validated by the robust Wald test ($F = 6.286$, $p = 0.000$). Short-term effects remain insignificant overall.

In summary, Cluster 3 represents a 'conflictual transition' that is characteristic of certain emerging African countries, where dynamic economic growth pushes towards a greener economy, but this is consistently hindered by rapid urbanisation and severe budgetary constraints.

Taken together, these results highlight three distinct energy transition pathways. Cluster 1 comprises structurally constrained economies where high public debt, an environmentally unfriendly growth model and a lacklustre financial system keep the share of renewable energies low, despite the presence of a stable long-term relationship. Cluster 2, on the other hand, comprises countries on a dynamic trajectory where economic growth and relatively controlled urbanisation are the main drivers of renewable energy growth, with financial and commercial channels playing a lesser role. Finally, cluster 3 illustrates a conflictual transition characterised by tension between efforts to achieve growth and diversification pushing towards a greener energy mix, and powerful structural constraints such as debt, unplanned urbanisation and an unfavourable macro-financial framework slowing down or diverting this process.

5 Conclusions

This study provides new insights into the diversity of Africa's renewable energy transition trajectories by developing a country typology based on macroeconomic, financial and structural characteristics. The results demonstrate that Africa's energy transition is not a homogeneous process, but rather a differentiated pathway shaped by countries' economic conditions, financial capacities and development constraints.

The main findings indicate three important patterns. First, countries facing weak economic fundamentals, high debt burdens and limited financial development encounter significant constraints in expanding renewable energy consumption, highlighting the importance of addressing broader development challenges before accelerating the transition. Second, countries with stronger economic performance, greater urbanisation and more developed financial systems display greater capacity to attract investment and integrate renewable energy into their growth strategies. Third, countries characterised by structural vulnerabilities require a balanced approach that combines economic reforms, improved governance and targeted investments to overcome persistent barriers to transition.

These findings underline that macroeconomic stability and financial development are essential foundations for successful energy transitions. Renewable energy policies should therefore be integrated within broader economic development strategies rather than implemented as isolated environmental initiatives.

From a policy perspective, the study suggests that differentiated approaches are required. For financially constrained countries, international support should prioritise concessional financing, debt sustainability measures and institutional capacity building. For countries with stronger transition potential, policies should focus on mobilising private investment through stable regulatory frameworks, financial innovation and infrastructure development. For countries facing structural bottlenecks, priority should be given to improving governance, enhancing public investment efficiency and promoting tailored energy solutions adapted to local conditions.

Overall, the success of Africa's energy transition will depend not only on technological advances and declining renewable energy costs, but also on the ability of policymakers and international partners to align climate objectives with economic realities. Recognising the diversity of national trajectories is therefore crucial for designing effective and inclusive strategies that transform the energy transition into a driver of sustainable development across the continent.

Author Contributions

Conceptualization, R.K. and C.G.; methodology, R.K.; software, R.K.; validation, R.K., C.G., and I.B.; formal analysis, R.K., C.G., and I.B.; investigation, R.K., C.G., and I.B.; resources, R.K. and C.G.; data curation, R.K. and C.G.; writing—original draft preparation, R.K., C.G., and I.B.; writing—review and editing, R.K. and C.G.; visualization, R.K. and C.G.; supervision, R.K. and C.G.; project administration, R.K. and C.G.; funding acquisition, I.B. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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